

Per and Polyfluoroalkyl Substances: Persistence, Exposure, and Accountability of Aqueous Film-Forming Foam Contamination

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**“ PFAS CONTAMINATION REPRESENTS A
SYSTEMATIC GOVERNANCE FAILURE
RATHER THAN SOLELY BEING A SCIENTIFIC
OR TECHNICAL PROBLEM. ”**

Abstract

Per and Polyfluoralkyl substances (PFAS) are a group of thousands of synthetic compounds that fail to decompose or break down. PFAS accumulate over time, causing a crisis of contamination in water sources, soil, and human bodies. PFAS are abundant in aqueous film-forming foam (AFFF), and the mass production and use of it create pollution/contamination in the immediate area of naval bases. When consequences ultimately occur, citizens look to their government for the assistance they're supposedly entitled to. However, recurring dismissal and failed remediation prove to be the sole outcome. The naval base, NAS Patuxent River (PAX), is compared and placed in the historical context of military pollution and government reaction efforts. This is done through using pre-existing scientific studies/results, selective policy and litigation review, and semi-structured interviews with various stakeholders. It has become apparent that remediation efforts fail due to factors like military immunity, the prioritization of corporate profit, and the weaponization of government ignorance. This places the burden of contamination on the victims. Moving forward, continuing this research is a vital step to understanding more effective paths of remediation

INTRODUCTION

Per and polyfluoroalkyl substances (PFAS) are detrimental to the environment and communities surrounding pollution hotspots. As forever chemicals that are widely used for firefighting purposes, PFAS have been added to aqueous film-forming foam (AFFF) formulas since their invention in the 1960s (Hu et al. 2016). PFAS can travel far past the initial site of release. Scholars in environmental science and public health have created extensive documentation on the toxicology and ecological distribution of PFAS. Research has definitively established that PFAS persists because of the strength of the carbon-fluorine bond, resisting hydrolysis, photolysis, and microbial degradation (Panieri et al. 2022; Rogers et al. 2025). As a result, these compounds bioaccumulate across trophic levels with documented tissue concentrations in fish livers reaching 8,700 times higher than surrounding water (Hill et al. 2024). The human health consequences include elevated risks for kidney, testicular, thyroid, and bladder cancers, ulcerative colitis, osteoporotic fractures, polycystic ovarian syndrome (PCOS), and immune suppression (ITRC 2025; Xu et al. 2021; Steenland & Winquist 2021). Those directly exposed to PFAS from occupational exposure show serum PFAS levels in the 99th percentile of the U.S. population (Santos 2026; Rosenfeld et al. 2023).

Concurrently, researchers in environmental policy and law have begun to examine the regulatory frameworks dictating hazardous substances and the ability to litigate areas like corporate and government liability. Despite clear evidence proving the toxicity of PFAS dating back to the 1960s, regulatory agencies have delayed action through the suppression of new scientific findings, weakening environmental and health standards, and prioritizing industry profit (Richter et al. 2025; Southerland 2026). The Environmental Protection Agency (EPA) was designed to operate as an independent entity, but has instead functioned as a barrier to remediation and accountability (Elder 2025). In addition, the Department of Defense (DOD) holds sovereign immunity and the Federal Facility Agreements that grant the military control over its own cleanup (Pietkoski 2020).

Legal doctrines like the Feres doctrine and the federal officer removal statute aid in protecting the military and corporate manufacturers from liability (Stokes 2023; Mystal 2025). At the state level, certain legislatures have implemented bans and disclosure laws, but that authority is undermined by federal preemption and Supreme Court precedent (Shook et al. 2025; Maryland Daily Record 2026). These existing policies create an illusion of protection while systematically harming public health.

There is extensive amount of scientific evidence and legislation because DOD sites have a history and trend of contaminating the surrounding communities. Naval Air Station (NAS) Patuxent River base (PAX) has been selected to compare with Camp Lejeune in North Carolina, and the 157th Air Refueling Wing of the New Hampshire National Guard. North Carolina and New Hampshire provide a historical pattern for the PAX base to be placed in. The selection of the PAX base is justified by the site's characteristics, not solely by geographic convenience. As a Superfund site since 1994, PAX has over three decades of regulatory remediation efforts, documentation, and contamination data (PATUXENT RIVER NAVAL AIR STATION). Moreover, the base's location on the Chesapeake Bay is of importance, as the downstream consequences of AFFF discharge are visible due to the Bay being an ecologically and economically vital estuary. PAX also shares the same legal and institutional structures that govern all DOD installations. This makes the findings from the PAX base generalizable to the various other military sites facing PFAS contamination across the country.

The dynamic of the interplay between the scientific process, policy formation, and regulatory implementation has yet to be adequately addressed in existing academia and research, especially in the context of military-grade contamination. Despite well-known case studies of military contamination, researchers must examine why similar contamination patterns cause weak, fragmented policy responses and public outcomes. There is a critical knowledge gap between academic fields, enabling the continuation of unabated contamination.

This study addresses that gap by examining four interconnected issues. First, the chemical persistence of PFAS and the documented impacts on human and biota health, supported by local biomonitoring data, toxicological studies from military sites, and ecotoxicological research on bioaccumulation in the Chesapeake Bay Web. Second, the institutional and legal structures that protect polluters, and the strategic production of ignorance through poorly designed and functioning government agencies. Third, the role of corporate manufacturers like 3M and DuPont, whose internal documents were revealed through litigation, showed concealed knowledge and the deliberate shifting of liability. Fourth, the environmental justice implications, which include the intentional and disproportionate siting of military bases and hazardous waste landfills in communities of color, and the burden of accountability that is placed on victims. Together, the four sections reveal the depths of systematic failures and provide a better understanding of why such failures persist.

An interdisciplinary study will be conducted to explore the contributors to systemic governance to understand the structural reforms necessary. It will analyze how scientific data on PFAS in marine life and water informs (or fails to inform) policy and litigation, evaluate the economic and enforcement failures of current laws, and interview individuals involved across a spectrum. This spectrum includes scientists in Southern Maryland, individuals a part of various communities suffering from PFAS contamination, and an environmental policy expert. The goal is to produce an integrated assessment that explains the current stagnation and multidisciplinary solutions for better environmental governance.

METHODS

This project utilizes a comparative case study approach, supported by mixed methods of data collection from multiple data sets and studies that range from official state-conducted work to unofficial, local work conducted by passionate individuals. Semi-structured interviews were conducted to understand and gain diverse perspectives. A selective

policy and judicial review provided the basis and sources for establishing the issues with current governing structures.

Engagement with Pre-Existing Data Sets

This research draws upon multiple scientific data sets and studies that employ different methodologies, spatial scales, and analytical targets. This approach reveals how methodological choices shape detection outcomes, consequently impacting regulatory responses. By consulting varied field studies, this research emphasizes methodological differences through data and results.

A local activist and scientist in Southern Maryland, Pat Elder, conducted targeted sampling near the PAX Webster Field Annex at several locations along the St. Mary's River, specifically areas directly downstream from the annex. Elder collected surface water, foam residue, oyster, crab, and rockfish tissues, and his own blood. These samples were collected using clean, non-fluorinated containers. Samples were analyzed using liquid chromatography-tandem mass spectrometry (LC-MS/MS) in accordance with EPA-approved PFAS testing methods (Elder 2024). This enabled detection at lower concentrations. Elder utilized a professional third-party laboratory for all PFAS analyses. The data set reflects its design to establish the presence of contamination.

The Maryland Department of the Environment (MDE) conducted a pilot study to examine PFAS in surface water and oysters near NAS Patuxent River. Systematic field sampling was conducted at multiple sites across the St. Mary's River and its tributaries during the summer months (Maryland Department of the Environment 2020).

The collected samples were tested for PFAS concentrations measured in parts per billion, and then compared to state screening levels and EPA drinking water advisories (Maryland Department of the Environment 2020). This government-led study used a broader spatial scale than Elder's, but set detection thresholds 1,000 times too high. This potentially understates contamination levels (Elder 2024).

The Maryland Watershed Association commissioned Eurofins Lancaster Laboratories for a PFAS study. The Eurofins study used EPA Method 537, Isotope Dilution, which is a targeted analytical method using LC-MS/MS to quantify trace levels of multiple PFAS compounds (Eurofins Lancaster Laboratories 2020). The lab used chain of custody documentation, preserved samples on ice, and high laboratory quality control (Eurofins Lancaster Laboratories 2020). This data set is important because it is independent from both state and military oversight, producing results that directly contradict MDE's conclusions.

Document Analysis

For policy and legal aspects, this research analyzes federal and state legislation, relevant judicial opinions, and issued regulatory documents. Sources include the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), the Freedom of Information Act (FOIA), the Toxic Substances Control Act (TSCA), and judicial opinions addressing sovereign immunity, the Feres doctrine, and the federal officer removal statute. State-level legislation from California, Colorado, Connecticut, Maine, Maryland, and Minnesota is examined to understand the divergence between state and federal regulatory action.

When employing legal analysis, the IRAC (Issue, Rule, Application, Conclusion) method was utilized. Each judicial opinion and statutory provision is narrowed to the core legal issue. The relevant rule of law, whether that is a statute, regulation, or precedent, is then identified. The rule is applied to the specific facts of PFAS contamination at military installations. A conclusion is then drawn about the legal outcome and its implications for accountability and remediation. This method ensures repeatable analysis of varying legal materials.

To document the perspective of communities impacted by military pollution, this research examines journalistic materials, including investigative reports by Sharon Lerner and Mariah Black. The local news coverage of contamination at NAS Patuxent River, Camp Lejeune, and the New Hampshire National Guard facility is included as well. These sources show the formation of relevant narratives, public engagement,

and the lived experiences of affected residents. Lived experiences are important in this research because such perspectives are often absent from typical academia.

This research also sought to obtain primary documentary evidence through the Freedom of Information Act (FOIA). FOIA is a federal law designed to provide public access to agency records, with the stated goal of government transparency. One FOIA request was filed with the Department of the Navy, and the second was filed with the Department of the Navy Archives. Both requests sought contracts between 3M and the Navy, as 3M was the original manufacturer of AFFF in the 1960s. This request was denied due to the absence of specific documents, and the denial indicated that the requested materials were stored at an archive department. The second FOIA request was filed with the appropriate archive department, but no response has been received. Under FOIA regulations and trends, a lack of response qualifies as a denial.

Semi-Structured Interviews

Four semi-structured interviews were conducted with stakeholders who have diverse perspectives on PFAS contamination. Pat Elder, a Southern Maryland scientist and activist living approximately 600 meters from the Webster Field Annex, provided scientific data and perspectives on government and military accountability. An individual working with the environment at PAX held an institutional perspective on the bases remediation efforts, sampling protocols, and operational constraints.

Doris Brock, whose husband died of cancer after 35 years of service in the New Hampshire National Guard, discussed her experience of ongoing advocacy work for veteran recognition and compensation. Another participant, M. Dibbles, holds environmental policy expertise and assisted in providing analyses of structural limits and environmental justice.

All interviews were recorded using OtterAI with prior informed consent from each participant. Institutional Review Board (IRB) approval was

received before any data collection. The PAX individual consented to the use of their name, but did not speak on behalf of any organization or institution, and especially not on behalf of the DOD or PAX specifically. Due to this, their name is removed along with their position. They are referenced as the PAX individual or PAX interviewee to distinguish their personal scientific experience from any institutional position, as they spoke on their perspective as an environmental scientist. Both Pat Elder and Doris Brock consented to the use of their names and requested no restrictions. M. Dibbles declined to have their name or affiliated work included and is therefore referred to as “M. Dibbles”.

The interviews follow a semi-structured format where questions varied by participant and did not strictly adhere to a pre-curated list. Questions were tailored to each stakeholder’s position and expertise. Content from certain interviews was deliberately included in subsequent interview questions to explore differing perspectives on the same issue. This includes the validity of independent testing and the role of scientific uncertainty in policy delay. No standardized list of questions was used, as the goal was to capture each participant’s framing of the problem rather than imposing a uniform structure.

The value of semi-structured interviews in environmental governance research lies in allowing researchers access to lived experiences, identifying various framings of complex problems, and understanding how stakeholders view causation and justice. Van Looy (2005) demonstrated how exploratory interviews with regulators, scientists, and project developers generated insights into regulatory gaps that desk-based analysis alone cannot identify. Similarly, Knill, Steinebach, and Zink (2025) conducted 157 semi-structured interviews with environmental agency members across six countries. This method allowed researchers to access information about implementation challenges and prioritization decisions. These are insights that quantitative data alone cannot provide.

LITERATURE REVIEW

The literature on PFAS contamination has grown exponentially over the past two decades. Environmental scientists and public health researchers have produced extensive documentation on the toxicology and bioaccumulation of PFAS compounds in both humans and wildlife populations (Panieri et al. 2022; Glaser et al. 2021; Baldwin et al. 2025). Concurrently, legal scholars and policy analysts have examined the regulatory frameworks dictating hazardous substances and the limits of immunity (Pietkoski 2020; Stokes 2023; Pollack et al. 2023).

The review is organized into sections that build toward the argument that PFAS contamination represents a systematic governance failure rather than solely being a scientific or technical problem. The beginning examines the chemistry and environmental behavior of PFAS, and how the compound's properties make it essential to aqueous film-forming foam (AFFF) (Panieri et al. 2022; Glaser et al. 2021; Hu et al. 2016). The following documents the human and ecological health consequences of PFAS exposure through ecotoxicological research, biomonitoring, and epidemiological studies (ITRC 2025; Elder 2024; Hill et al. 2024; Hoverman et al. 2025; Holden et al. 2025). The historical pattern of military contamination is then established through well-documented cases, along with the corporate conduct of 3M and DuPont (Bove et al. 2024; Brock 2026; Richter et al. 2025; Bilott 2019). The examination of regulatory and legal work then demonstrates how laws function as barriers to accountability (Pietkoski 2020; Stokes 2023; Mystal 2025). Together, these sections show that the persistence of PFAS in the environment is mirrored in the persistence of institutional delays, regulatory capture, and doctrines that prioritize profit over justice.

PFAS Chemistry and Environmental Behavior

The chemical and molecular properties of per- and polyfluoroalkyl (PFAS) compounds are the reason it has become indispensable to military firefighting foams. The bond that creates environmental persistence also produces the thermal stability and surface tension reduction required for aqueous film-forming foam (AFFF).

Understanding the chemistry of PFAS builds the foundation for why military installations adopted and continue using these compounds. AFFF is designed to extinguish high-hazard Class B 16 flammable liquid fires (like fuel, oil, and solvents) by creating a thin, aqueous film that spreads quickly across the surface of the fuel, which smothers the fire and flammable vapors (Hu et al. 2016). This aspect of the formula is due to the fluorosurfactants (PFAS-based compounds). Surfactants are molecules that have both a hydrophobic and a hydrophilic part. In a fluorosurfactant, the hydrophobic tail is a fluorinated chain, and the extreme strength and low polarity of the C-F bond create a repulsion of both water and oil. The dual repellency is crucial because the fluorosurfactants significantly lower the surface tension of the water, allowing the foam to easily spread across the surface of a low surface tension fuel like gasoline. It also provides great chemical and thermal stability, as it ensures the suppressant blanket remains effective under the heat of a fire instead of degrading upon contact with the fuel (Hu et al. 2016).

Known as forever chemicals, the persistence of PFAS is attributed to the carbon-fluorine (C-F) bond. Fluorine is the most electronegative element on the periodic table, meaning it has an extremely strong attraction for electrons (Panieri et al. 2022). When fluorine is bonded to carbon, the high electronegativity makes a very polarized and strong bond. Its strength as a function of its polarity is reinforced by the small size of the fluorine atom, allowing it to have optimal orbital overlap with the carbon atom (greater overlap increases the stability and strength of a bond), and specific electrostatic attractions that contribute to the bond's stability.

The powerful bond strength is a direct reason for its resistance against environmental degradation processes. The carbon fluorine bond resists hydrolysis (a chemical process in which water reacts with a compound, breaking chemical bonds), photolysis (another chemical process where ultraviolet radiation causes chemical bonds to break apart), and microbial degradation which is the breakdown of substances by

microorganisms like bacteria or fungi, using enzymatic processes to transform or mineralize compounds (Rogers et al. 2025; Li et al. 2025; "Photolysis" 2026; Trigo et al. 2009; "Biodegradation" 2026; Shane 2019). Hydrolysis is insufficient for proper and full degradation of the C-F bond because the electron density and steric hindrance (prevention of chemical reactions by large groups of atoms blocking the reaction site) by the fluorine atoms protect the carbon structure (Khan et al. 2019; Laçın et al. 2023). Similarly, the energy from ultraviolet radiation is usually not enough to break the C-F bond (Steinberg & Paul 2008; Sharma et al. 2022). Microorganisms, although adept at degrading most natural and synthetic organic compounds, lack the enzymatic ability necessary to degrade the perfluorinated bonds. The energy required to break the bond is so high that microbial metabolism offers no evolutionary advantage, leaving the compounds intact in environments for decades, and even centuries (Skeel & Boreen 2020; Sadowsky et al. 2014).

PFAS are put into two main categories, perfluoroalkyl substances and polyfluoroalkyl substances, based on their structure (Glaser et al. 2021). In perfluoroalkyls, all hydrogen atoms on a continuous carbon chain are replaced by fluorine atoms, which makes them fully fluorinated. This group includes the most well-known and studied compounds, such as PFOs (perfluorooctane sulfonic acid) and PFOAs (perfluorooctanoic acid). Polyfluoroalkyl substances are not fully fluorinated but contain a mix of fluorinated and non-fluorinated carbons along their chain. This difference in structure means that polyfluoroalkyl substances can act as precursor compounds (substances in a chemical reaction that produce another compound), eventually forming the more persistent perfluoroalkyls previously mentioned (Glaser et al. 2021)

The PFAS family is further categorized by chain length, as it dictates their transport. Long-chain PFAS are defined by the group attached to the fluorinated tail. PFCAs (perfluoroalkyl carboxylic acids) have seven or more fluorinated carbons, and PFSAAs (perfluoroalkyl sulfonic acids) have six or more fluorinated carbons.

PFOAs (a C8 PFCA) and PFOs (a C8 PFSA) are long-chain compounds and are relatively less water-soluble. Consequently, they tend to attach to soils, sediments, and sludge, accumulating on solids and migrating slowly underground, creating higher concentrations of contamination in the immediate surrounding area of the initial site of release (Malik et al. 2024). Short-chain PFAS are defined as PFCAs with fewer than seven carbons and PFSAs with fewer than six carbons. Due to their shorter and less hydrophobic fluorinated tail, they are significantly more water-soluble, less likely to adsorb to soil particles, and highly mobile in the environment, allowing them to travel quickly with water currents, seep into groundwater, and contaminate water sources far past the point of release (Malik et al. 2024). This creates an issue as they are harder to contain for remediation efforts.

Historically, AFFF formulas were primarily long-chain PFAS, mainly PFOs and their precursors. Due to the global discovery of their prevalence and toxicity, production shifted in the early 2000s to fluorotelomerization-based formulations, which produced short-chain PFAS (referred to as C6-AFFF) as replacements (Travis 2024). As a result of this, the chemical makeup of AFFF contamination at military sites is complex. It can consist of the original long-chain compounds from older formulas, the short-chain replacements from modern foams, and a plethora of polyfluoroalkyl precursor compounds belonging to fluorotelomer-based foams, which slowly turn into terminal compounds. So, older C8 (PFOA or PFOs) based AFFF created through electrochemical fluorination were phased out for C6 foams and designed to break down through time (Travis 2024). However, through their degradation path, they become precursors and result in the formation of terminal compounds. The early discovery of the presence and danger of PFAS led to some scientific advancement and change, but there is a clear failure in genuine remediation results when only utilizing technological inventions.

The use of AFFF during firefighting emergencies and routine training exercises brings a complex mix of PFAS directly into the environment.

Long-chain compounds are retained in the upper layers of soil, creating a long-term source of contamination, while short-chain PFAS rapidly leach through the soil column and act as an immediate contaminant to groundwater. Once in the groundwater, it can discharge into surface water bodies such as streams, rivers, and estuaries (Khan et al. 2023). The ratio of different PFAS compounds can serve as a "chemical fingerprint" that reveals transport pathways and the ongoing transformation of the original AFFF mixture as it remains in the environment. Furthermore, as PFAS cycle through food webs, bioaccumulation can vary per species and ecological niches, with higher concentrations found in predatory species and organisms with longer lifespans (Glaser et al. 2021).

Human and Ecological Health Consequences

The common practice of using aqueous film-forming foam has resulted in severe, lasting consequences for both human and ecological health. As a critical public health concern, Baldwin et al. (2025) emphasize the persistence of PFAS as the greatest concern, as the contaminated sites require fiscally extensive remediation strategies and pose chronic toxicological risk with transgenerational consequences. Moreover, the transition to fluorine-free foam (F3) alternatives addresses certain concerns, but it introduces additional uncertainties. New research has found that some replacement formulations may be more acutely toxic to aquatic species than the original formulas (Hold et al. 2025), and the long-term ecological impacts remain poorly understood. As concluded through several studies, there is a need for continued research to inform proper regulatory decisions and remediation strategies that will adequately protect human communities and the ecosystems they depend on.

Human Health Consequences

The human health consequences of AFFF-based PFAS exposure have been mainly documented through epidemiological studies of communities with contaminated drinking water.

The most extensive study so far takes place in Ronneby, Sweden, where AFFF from a nearby military base contaminated the municipal water supply from the mid-1980s until 2013 (ITRC 2025). Biometric monitoring of approximately 3,300 residents of Ronneby showed that the geometric mean serum levels (used to determine the central tendency of biomarker concentrations) of PFHxS (perfluorohexane sulfonate), PFOs, and PFOA were about 135x, 35x, and 5x higher than levels in a reference population (Xu et al. 2021). These levels of exposure in Ronneby's population gave crucial insight into the health issues of chronic PFAS exposure.

Cancer risks represent a primary concern when discussing PFAS exposure. Rosenfeld et al. (2023) state that international, national, and state agencies have concluded that PFOA is potentially carcinogenic, and studies on cancer risk among occupationally exposed firefighters show an increased risk for thyroid, bladder, kidney, testicular, prostate, and colon cancer (Li et al. 2022). The Ronneby cohort study found evidence for an elevated risk of kidney cancer with a non-statistically significant hazard ratio of 1.85 for kidney cancer in individuals with the highest exposure rates compared to those with lesser rates or no exposure (IRTC 2025). The authors also observed statistically insignificant increases in hazard ratios for bladder, brain, bone and cartilage, thyroid, and testicular cancers. It is important to note that although the hazard ratio of 1.85 is considered statistically insignificant, it aligns with findings from larger studies, such as the C8 Health Project and a meta-analysis by Steenland and Winquist (2021), which identified a relative risk of 1.27 for kidney cancer associated with PFOA exposure. When several studies show elevated, individually non-significant estimates in the same direction, the collective evidence creates a causal inference that PFAS do have a relationship with the development of certain cancers.

Beyond this, PFAS exposure has also been linked to other adverse health issues.

The research on Ronneby has documented significantly increased risks of major osteoporotic fractures (bone breaks caused by weakened bone density) with a hazard ratio of 1.11 and hip fractures (HR = 1.12) among those with the highest exposure levels (Xu et al. 2023). Hammarstrand et al. (2021) found that higher PFAS exposure was linked with significantly elevated hazard ratios for polycystic ovarian syndrome (PCOS, a hormonal disorder) (HR = 2.18). However, there were no associations found for endometriosis or uterine leiomyoma (ITRC 2025). There were mixed findings for thyroid issues, as Andersson et al. (2019) found no increased risk of hyperthyroidism (an overactive thyroid), and Li et al. (2021) noted no consistent evidence that links PFAS ingestion to alterations in thyroid hormone levels. The variety and extent of impacts shown through this data emphasize the importance and difficulty of PFAS toxicology.

The level and consistency of PFAS exposure dictate the severity and likelihood of health consequences. This leads to vulnerable populations, like minority communities or certain occupational fields, being at a higher risk. Firefighters represent a highly vulnerable group due to direct exposure to AFFF during training and emergency response tasks. Rosenfeld et al. (2023) documented that firefighters have been found to have elevated serum levels of long-chain PFAS, and because of bioaccumulation in the body, firefighters experience cumulative effects of exposure. Recent research involving more than 2,000 firefighters from seven states found that every individual had detectable PFAS in their blood. There were especially elevated levels found in those associated with original AFFF formulations, prior military service, and reliance on well water as a primary drinking source (Santos 2026). A Southern Maryland activist, scientist, and long-term citizen of St. Mary's, Pat Elder, showed that even individuals with non-PFAS exposure occupations have elevated levels of PFAS likely due to residing near a naval base. Elder resides in the Chesapeake Bay, about 600 meters across a creek from the Webster Field Annex of the Patuxent River Naval Air Station, where the Navy has conducted regular firefighting training exercises using AFFF for decades. Elder's personal biomonitoring

results, published in November 2024, documented extraordinarily high serum PFAS levels that place him in the 99th percentile of the U.S. population.

From washed-up shore foam that contains PFAS to the oysters he would harvest from the docks, a part of his home property, Elder regularly interacts with water sources that have been contaminated from naval exercises. His consumption of oysters, crabs, and rockfish as a typical Chesapeake Bay resident represents an example of PFAS bioaccumulating through the aquatic food web. Elder's blood serum testing was done by Eurofins Environmental Testing and found a total PFAS concentration of 42.2 ng/mL (nanograms per milliliter, equivalent to parts per billion) for the seven PFAS compounds identified by the National Academies of Sciences, Engineering, and Medicine as clinically significant. Astoundingly, his PFOS levels alone measured 32.3 ng/mL

In the context of the Agency for Toxic Substances and Disease Registry (ATSDR), the United States average PFOs level is 4.25 ng/mL, and 95% of the population has levels of 14.6 ng/mL or lower (only 5% have more than 14.6 ng/mL because PFAS concentrations in the general population are right-skewed). Elder's concentration of 42.16 ng/mL puts him far above the 20 ng/mL threshold that the National Academies use to encourage enhanced clinical screening, meaning Elder has a substantially increased risk for issues with thyroid function, kidney and testicular cancer, and ulcerative colitis (chronic inflammatory bowel disease). This consequence is placed upon an individual, a part of the regular general population, meaning those near military installations are likely unknowingly consuming PFAS contamination at harmful levels.

Impact on Wildlife Health

Along with the concern for human health, the ecological consequences of AFFF-derived PFAS are equally concerning due to documented bioaccumulation across 25 various biota.

The chemicals' stability and tendency to accumulate in biological tissues provide the foundation for assessing PFAS impacts on wildlife. As explained in Baldwin et al. (2025), the C-F bonds that enable PFAS' fire-suppressing properties also allow them to resist environmental degradation, creating long environmental half-lives that build chronic toxicity.

The ecotoxicological significance of AFFF and its replacements has impacted aquatic invertebrates that form the foundation of freshwater food webs. Research on a keystone zooplankton species, the *Daphnia magna*, found that fluorine-free foam alternatives can be more acutely toxic than the traditional PFA-based foams. Through a multi-generational exposure study, Holden et al. (2025) identified that exposure to three of six F3 alternatives led to changes in tolerance in the exposed populations in comparison to unexposed controls, with two populations displaying increased tolerance (potential adaptation) and one population showing decreased tolerance (indicating cumulative stress). The authors discussed these results as finding that the replacement formulations were more acutely toxic to aquatic species than the PFAS-based AFFF they were designed to replace. This is significant for regulatory decisions because it demonstrates that PFA-free alternatives pose their own ecotoxicological risks and that replacements are not to be utilized as single solutions.

Amphibians, which occupy an important ecological niche that bridges aquatic and terrestrial ecosystems, are particularly vulnerable to aqueous film-forming foam. Hoverman et al. (2025) studied the impacts of both PFAS-based AFFF, as well as fluorine-free foam alternatives (F3), on the larval development of the gray treefrog (*Hyla Versicolor*) species. Their study found that the PFAS-based AFFF formulation from Buckeye Platinum Plus reduced survival and delayed metamorphosis at the highest concentration that was tested. In comparison, the F3 alternatives showed comparatively limited effects on the completed transformation endpoint (Hoverman et al. 2025).

However, the study's behavioral observations found that tadpoles exposed to several foam formulations (including PFAS-based) sought refuge more frequently, resulting in 19 to 30 percent fewer tadpoles observed in open water compared to the controls (Hoverman et al. 2025). Although the behavioral shift did not directly alter or affect growth and development, such changes can negatively affect ecological interactions, like predation, parasitism, and competition, potentially impacting population-level outcomes.

Fish species are at a higher trophic level and are bioaccumulators. Fish exposed to AFFF contamination demonstrate substantial accumulation with specific tissue patterns. A controlled exposure study conducted by Hill et al. (2024) used fathead minnows exposed to PFAS-contaminated groundwater from Joint Base Cape Cod, Massachusetts (a naval site with decades of AFFF usage). It was found that the total PFAS concentrations in fish tissues increased over the course of 21 days, and the accumulation followed the distinct pattern of liver > kidney > gonad (Hill et al. 2024). The liver had the highest measured amount, the kidney had intermediate concentrations, and the gonad had the lowest concentration of the tested tissues. The tissue distribution matters because it indicates that PFAS accumulate in metabolically active (cells, tissues, or organisms engaged in biochemical processes to sustain life and function) and reproductive organs. This holds potential implications for metabolic function and reproductive success. On a concerning note, the study reported bioconcentration for perfluorooctane sulfonamide (FOSA), reaching 8,700 L/kg in the kidney tissue. This means that the PFAS compound becomes 8,700 times more concentrated in the fish's kidney than in the water it swims in. Another simple comparison to emphasize the power of accumulation is if 8,700 liters of water, enough to fill about 45 bathtubs, were to be condensed into a single organ the size of a marble (note that the size of the organ does not automatically produce a high bioconcentration factor; 8,700 L/kg is the normalized value).

An additional concern in the study's findings is that precursor compounds were present in both the groundwater and fish tissues. Hill et al. (2024) identified perfluoroalkyl sulfinates and perfluoroalkyl sulfonamide compounds in the liver tissue, which indicates the presence of alternative pathways for terminal PFAS exposure in aquatic wildlife. This is important because it shows that even when terminal PFAS concentrations are addressed and properly managed, precursor compounds can continue to exist as hidden reservoirs that ultimately contribute to the persistence of PFAS foams.

Another sphere of consequences is the avian species inhabiting areas adjacent to AFFF-contaminated sites. A study of tree swallows (*Tachycineta Bicolor*) near AFFF sources in Duluth, Minnesota, found that the total PFAS concentration in eggs at the National Guard Base site was 30 to 40 times higher than at a reference lake, and nestling PFAS concentrations were 10 to 15 times higher (Custer et al. 2024). The study identified PFHxS as a distinctive marker of AFFF contamination, and it comprised a higher proportion of total PFAS at AFFF-contaminated sites compared to reference locations or sites with primarily urban and industrial PFAS sources (Custer et al. 2024). Despite the substantial exposure differences, the researchers were unable to connect biomarker responses with PFAS exposure levels. This creates a complex challenge because there is a disconnect between documented bioaccumulation and observable physiological effects, further complicating risk assessment and management efforts.

In addition to the mere existence of PFAS in wildlife, there is a concern about the transgenerational consequences that may extend beyond the organisms directly exposed. A multi-generational study conducted by Holden et al. (2025) on *Daphnia* (small freshwater crustaceans) is the first look at evolutionary responses to fluorine-free foam alternatives, which resulted in demonstrating that chronic exposure over 84 days (about 10 to 12 generations for the species) can ultimately alter population tolerance.

Although the idea of evolutionary change in this issue may sound positive, when organisms divert their energy toward detoxification and survival, it may come at the expense of other functions.

Evolutionary trade-offs include reduced reproductive output (fewer or smaller offspring and reduced fertility), decreased competitive ability (the tolerant species may now only fit in to a contaminated environment), increased vulnerability to other stressors (a major issue in the face of climate change), genetic erosion and loss of genetic diversity, and increased population fragility, all of which can result to overall ecological collapse as *Daphnia* are a keystone species. Epigenetic inheritance is an additional consequence, as PFAS are endocrine disruptors that potentially alter the epigenetic marks that control which genes are turned on or off. Altered epigenetic states can cause disease and developmental issues, which will then impact unexposed generations, prolonging and further complicating mitigation and overall scientific understanding (Kebieche et al. 2025; Stefani et al. 2014; Holden et al. 2025; Cunha et al. 2025; Svoboda et al. 2022). The evidence across trophic levels undeniably shows that PFAS contamination from AFFF does not remain confined to the point of release. Since the scientific evidence makes clear that AFFF contamination exists, the focus must shift from occurrence and presence to how it occurs. More specifically, how military installations have used, disposed of, and continued to release AFFF for decades.

Military Use of Aqueous Film-Forming Foam and Evidence of Contamination

Military installations are primary PFAS contamination hotspots due to the decades of reliance on aqueous film-forming foams (AFFF) for firefighting training and emergency response. Unlike the industrial sources that are, to some degree, subject to state oversight, military facilities have operated under protocols that prioritize mission objectives without the proper understanding of or a cleanup plan for environmental contamination.

This has resulted in the common practice, and often unmonitored, release of PFAS foam into soils, groundwater, and surrounding waterways (Pietkoski 2020). Such practices have created an environmental legacy that is detrimentally affecting municipal water supplies and downstream ecosystems. Some literature has established that AFFF use is a “predominant point source of PFAS contamination...” which includes primary precursors, secondary precursors, and perfluoroalkyl acids (Yan et al. 2024).

An important focus on specific military AFFF PFAS is exemplified through a comprehensive review completed by Yan and colleagues. Yan et al. found significant discrepancies between the initial PFAS concentration profile in AFFF formulations and the concentration profile measured in AFFF-contaminated environments. Primary precursors make up 52.6 percent of PFAS mass in electrochemical fluorination (ECF)-based AFFF, and 99.5 percent of PFAS mass in fluorotelomer (FT)-based AFFFs, yet they represent only 0.7 percent of the total mass in AFFF-contaminated groundwater. In contrast, secondary precursors, which are less than 1 percent of PFAS in AFFFs, are 4.0 to 27.8 percent of PFAS in AFFF-contaminated environments (Yan et al. 2024). This difference is due to in situ biotransformation processes (indigenous microorganisms in soil/groundwater breaking down organic compounds), which means the contaminants found in today's environmental samples are not quite the same compounds that were originally released, further complicating source attribution and toxicity assessment. Understanding transformation dynamics is necessary but not sufficient for explaining the entire scope of contamination. To understand what quantities, through what disposal methods, and with what level of institutional awareness, the historical practices must be examined.

Historical Disposal and Contamination

The historical practices that created this contamination pattern are well-documented in regulatory records and firsthand accounts.

An environmental scientist who has experience with NAS PAX River base practices was interviewed, and they described the routine disposal methods that were standard across military bases before environmental regulations existed. Personnel would discharge AFFF onto the ground to test equipment or dispose of expired foam by pouring it out beside buildings, both of which were noted as “pretty consistent at all the installations” 1 . Such discharges were not considered problematic because the environmental risks associated with PFAS were not yet established. The result was the creation of predictable contamination zones near training spaces, hangars, and equipment storage sites - locations that now require investigation and remediation. When AFFFs are released during training exercises, firefighting operations, or accidental spills, PFAS bind to surface soil particles and rest in the upper layers of the subsurface. The foam can also infiltrate further into the subsurface, resulting in the formation of PFAS-contaminated groundwater plumes, which can discharge to surface water

Moreover, the foam releases can become runoff that brings PFAS to nearby surface water bodies, where the compounds have the chance to accumulate in sediment and infiltrate the food chain through bioaccumulation (Yan et al. 2024). Some AFFF-derived PFAS can become airborne and transported by air currents to be deposited onto surface soils and sediments via atmospheric deposition (transfer of particles from the atmosphere to the surface via precipitation or gravitational settling), creating contamination at far-removed distances from the original point of release (Yan et al. 2024).

PFAS Composition in Military Soils

Recent research quantified PFAS mixtures in surface soils at military installations to provide necessary data for ecological risk assessments. East et al. (2025) used a formalized prioritization method on a large dataset of PFAS concentrations in surface soil at AFFF-contaminated sites on active and former U.S. Air Force installations. They found that perfluorooctane sulfonate (PFOS) occurred most commonly and at the highest concentration across the PFAS measured.

It was also found that two to three PFAS made up 86 to 91 percent of the sum of PFAS in any given site-specific mixture. After PFOs, the most common and highest concentrated PFAS in order were PFHxS, PFOA, PFOSA, and PFHxA (East et al. 2025). Site-specific PFAS mixtures were found to be approximately 5 to 12% PFHxS, PFOA, PFOSA, and PFHxA, with PFOS comprising approximately 82% of the total PFAS mass. This profile holds important implications for risk assessments and remediation, as the dominance of PFOs in military installation soils, despite being phased out of ECF-based AFFF formulations after 2002, shows the sheer persistence of this compound. In addition, the consistent finding of two to three PFAS compounds constituting the vast majority of contamination at any given site implies that target remediation and risk assessment efforts could focus on a small number of priority compounds while still addressing the majority of environmental risk.

The uncertainty identified by East et al. (2025) has direct implications for PAX. The inconsistent prevalence of fluorotelomer sulfonates across military sites means that contamination profiles at PAX may differ from those at other bases. This would depend on which AFFF formulations were used and when they were released. Without knowing the specific mixture present in nearby soils and groundwater, risk assessments may target the wrong compounds. Unknown or poorly understood toxicological potency complicates efforts to determine whether PAX's contamination levels pose a threat to human and ecological health. These findings show that relying on a narrow set of priority compounds, similar to both the MDE and the base's own monitoring, systematically understates risk.

Contamination at Naval Air Station Patuxent River

NAS Patuxent River, labeled a Superfund site since May 31, 1994, is an example in Southern Maryland that illustrates the complexity of PFAS contamination assessment and the institutional tensions that occur when government and independent findings diverge.

The base, established in 1943, has a long history of AFFF use dating back to its invention in the 1960s, where firefighting training areas, hangars, and aircraft maintenance facilities serve as primary release points. The PAX individual interviewed for this research noted that the base has taken proactive steps under the Installation Restoration Program, including the early adoption of PFAS sampling and engaging with state and federal regulators through a Restoration Advisory Board that includes public representatives.

Despite the scientific discrepancies, NAS Patuxent River has taken efforts to address PFAS contamination under the Installation Restoration Program. The PAX interviewee discussed the base's proactive approach that puts PAX ahead of many other bases in responding to PFA's concerns. The base started requesting funding for PFAS investigation before the Department of Defense (DOD) formally established PFAS-focused programs. PAX became "one of the first Department of Defense installations to get out there and start doing investigative sampling" ². This proactive initiative allowed the base to build an understanding of contamination through archives that included interviews with retired firefighters and long-term personnel who could identify the historical disposal and practice locations.

PAX's Restoration Advisory Board (RAB) includes representatives from the Maryland Department of the Environment (MDE), the Environmental Protection Agency (EPA), St. Mary's County government, civilian community members, and local tribal nations. The RAB functions under a formal Federal Facilities Agreement that mandates transparency and public participation, and all sampling plans are reviewed by state and federal regulators before implementation. According to the interviewee, such a collaborative structure has created the balance necessary for thorough investigation with fiscal responsibility. All of their sampling results are made public and are available through the base's website, at the local library, and during public RAB meetings.

Those meetings have included efforts to increase community participation through catered meals and outreach to local organizations, showing the intention to include and educate the public on the reality of the contamination and current efforts, something contrary to notions made by critics.

However, the base has faced challenges with remediation due to institutional and technological limitations. The individual discussed the difficulty disposing of PFAS-contaminated materials due to the lack of treatment technologies. When the DOD took the stance of treating PFAS as hazardous waste (even before being formally listed as such under federal regulations), the base found that no landfills would or could accept PFAS-contaminated soils or water. This created operational challenges, as construction projects were unable to proceed due to indisposible excavated contaminated soils. If the base stockpiled the material, there would be new contamination sites as a result. PAX eventually secured a contract with a hazardous waste landfill in Emelle, Alabama, the only site willing to accept the waste.

Due to disposal issues, PAX has pursued alternative treatment technologies and disposal methods. The base has piloted thermal treatment projects to destroy PFAS compounds and leaching processes designed to separate PFAS from contaminated soils. Recently, the DOD has permitted disposal at Subtitle D landfills, which are standard landfills equipped with liners and leachate collection systems, provided that the landfills are willing to accept PFAS waste. However, as noted by the interviewee, most landfills remain reluctant to accept PFAS materials, showing the opposition of NIMBY (Not In My Backyard) from communities near proposed disposal sites. The base's efforts are supported through congressionally appropriated restoration funds, which are separate from base operating budgets, meaning that the cleanup does not compete with other priorities or problems for funding. Such dedicated funding, the individual stated, has allowed the restoration program to continue even during "austere years" when other infrastructure projects received no discretionary funding.

Potentially most significant, the interviewee described a shift in the base's approach to AFFF management. After 45 to 50 years of storing AFFF in hangars without a single documented fire requiring its use, the base decided to remove the AFFF systems and replaced it with water mist systems, and limit foam to cash trucks for emergency response. According to the individual interviewed, this decision was a risk-benefit choice - the infrastructure loss from a fire was easier to replace than long-term remediation costs and liability of AFFF storage and potential release. The base has also made the decision not to transition to F3, as the individual cited the new evidence that the alternatives are toxic to the food chains in the Bay, showing the level of responsiveness the base has to new scientific findings, even when it complicates transition timelines.

Aside from any measures taken, the empirical record of PFA detection at Patuxent River has significant contradictions between official and independent assessments. In 2020, MDE conducted a pilot study examining PFAS in surface waters and oysters in the St. Mary's River system. MDE reported no PFAS detections of concern and concluded that contamination levels did not warrant additional investigation or public health advisories (Maryland Department of the Environment 2020). These findings were used to reassure local communities and justify a limited regulatory response. The MDE based its study on detection thresholds that critics argue were too high to capture ecologically meaningful contamination, and the sampling strategy underestimated spatial and temporal variability in PFAS distribution (Elder "MDE Report Deceives Public").

Independent analyses commissioned by the Maryland Watershed Association produced different conclusions. Eurofins Lancaster Laboratories detected PFAS in aquatic organisms that are consistent with bioaccumulation and chronic exposure, which directly contradicts the MDE's reported findings (Eurofins Lancaster Laboratories 2020).

The Eurofins report documented PFAS concentrations in fish and shellfish samples taken from the St. Mary's River, that infers current contamination in the aquatic food web. These findings were especially concerning due to the ecological and economic importance of the St. Mary's River System, as it supports recreational and commercial fishing, and the broader Chesapeake Bay ecosystem it flows into.

The individual interviewed acknowledged the challenges with methodology that were behind the discrepancies and stated that early PFAS testing faced significant limitations. When the EPA first began requiring PFAS testing, "there were no labs that could meet what EPA in theory wanted," and many facilities that claimed to test for PFAS did not specify which of the thousands of chemical compounds they were detecting⁴. The lack of a standardized PFAS sampling protocol produced inconsistent results, but recent guidance from agencies such as the Michigan Department of Environment, Great Lakes, and Energy (2022) has improved comparability and reliability. Regardless, discrepancies persist, which raise questions about how scientific uncertainty is interpreted and mobilized in government decision-making. However, strong criticism of the MDE study remains.

Pat Elder, the local scientist and activist with high serum PFAS levels, continued the discussion of the MDE results. Elder stated that the sampling strategy relied on a limited number of collection points during a single season, failing to account for tidal dynamics, which may move contaminants into areas not sampled by the state. Moreover, in December 2024, Elder reported his discovery of PFAS foam residue on the beach at St. Mary's College of Maryland, which is located ten miles downstream from the base (Elder 2024). The presence of foam residues suggests current or recent releases of AFFF into the river, contradicting official assurances that contamination was historical, limited, and contained.

The difference between MDE's and Eurofin's findings shows a broader pattern in the literature.

Proffitt (2022) documents the evolution of sampling protocols, finding that the absence of standardized methods historically contributed to the inconsistent results across studies. When government studies report no cause for concern while independent studies have a polar opposite report, the subsequent confusion further erodes public trust, delays genuine remediation, and allows for the persistence of contamination while responsibility is constantly disputed among agencies.

Establishing a Pattern of Contamination: Historical Context

The contamination at Patuxent River cannot be analyzed alone, as it is part of a larger historical pattern of military pollution for which Camp Lejeune can represent due to its extensive documentation and presence in the media. The Camp Lejeune Marine Corps Base in North Carolina had a water contamination issue that persisted for decades before federal agencies acknowledged health risks or implemented remedial action. The contamination was a result of the disposal of industrial solvents, which include trichloroethylene (TCE), perchloroethylene (PCE), and benzene, into the base's water supply that supplied housing, barracks, and administrative buildings from the 1950s through the 1980s (National Academies Press 2009). The evidence from Camp Lejeune linked military installation water contamination to dire health outcomes. A cohort study by Bove et al. (2024) researched cancer among Marines, Navy personnel, and civilian workers exposed to the contaminated drinking water. The study found significantly elevated incidence rates for several cancers among exposed populations. Marines and Navy personnel had higher rates of acute myeloid leukemia, all myeloid cancers combined, cancers of the esophagus, larynx, soft tissue, and thyroid. Civilian workers had increased hazard ratios for lung cancer, female breast cancer, and all myeloid cancers (Bove et al. 2024).

A related study conducted by the same research team examined mortality in exposed populations and found increased mortality rates for several causes of death.

It documented increased mortality from cancers of the kidney, esophagus, and liver, along with Parkinson's disease and amyotrophic lateral sclerosis (ALS) among the groups exposed to contaminated water at Camp Lejeune (Bove et al. 2024). The consistency of those findings strengthened the causal inference linking military water contamination to detrimental health outcomes. In sum, Bove's research is the most epidemiological investigative study of military water contamination to date. The research team accessed the DOD's personnel records, reconstructed historical water contamination levels, and tracked health outcomes over decades. The finding on civilian workers, who were not subject to the same chain of command and typically lived off base, experienced elevated cancer rates, reinforcing the conclusion that the contamination was responsible for the 41 observed health effects, reducing the ability to deflect blame on genetics, misfortune, or other parts of military service.

The Camp Lejeune contamination established a form of precedent, in a non-legal context, that has several implications for other military bases, including PAX. Camp Lejeune demonstrated that institutional delays in the acknowledgement of contamination and implementation of remediation can span decades, during which exposed individuals will continue to suffer health consequences. It showed the limitations of the United States' governing and regulatory systems that enable military institutions to operate and pollute under exemptions. Lastly, it showed that once contamination has received acknowledgement, establishing causation and compensating affected populations can be contentious and non-existent, especially as many exposed individuals die before receiving recognition or compensation.

Although Camp Lejeune is a notorious case of military pollution, there are many other bases with similar or worse stories and data. The story of Doris Brock, whose husband served full-time in the New Hampshire National Guard, solidifies the historical and nationwide pattern of harm caused by military installation contamination and the institutional failures that allow and worsen it.

Brock's husband, Kendall Brock, served 35 years with the 157th Air Refueling Wing of the National Guard and was diagnosed with stage three bladder cancer in 2015, which rapidly progressed to stage four. Kendall Brock died in 2017. In the years following her husband's diagnosis, Doris Brock began documenting the health outcomes of her husband's colleagues, all of whom had served alongside him at the same base, performing the same jobs, for decades. Out of 66 friends, 34 died either before or shortly after retirement. "That number was like, very, very large to me," Brock stated, "for a small group of people, because the New Hampshire National Guard is not as big as the California National Guard" 5 .

Brock's experience follows the same pattern documented at Camp Lejeune. As Doris Brock's story gained traction in the media, a Marine Corps veteran who became one of the most prominent voices for Camp Lejeune families after his nine-year-old daughter died of leukemia linked to contaminated water, Jerry Ensminger, reached out. Ensminger warned Brock that she was to face "a horrible fight" when demanding accountability and justice, which is exactly what happened. The Camp Lejeune litigation spanned about four decades before Congress finally passed the Camp Lejeune Justice Act in 2022, but even then, the compensation process has been halted by denials and bureaucratic obstacles (National Academies Press 2009). As Brock stated, "almost 40 years worth of fighting, and they really aren't much further ahead than the day they started, other than the fact that everybody knows who Camp Lejeune is."

Both Camp Lejeune and the New Hampshire National Guard facility experienced delays that have lasted for decades. The New Hampshire site was designated as a Superfund site in 2014 due to industrial solvents, degreasers, and other aircraft repair chemicals, all of which Brock noted to predate the current focus on PFAS. When she began her advocacy, PFAS "wasn't even a buzzword" 6 .

Her fight focuses on the solvents and chemicals used in military operations, many of which are well-known carcinogens with established health consequence profiles (Agency for Toxic Substances and Disease Registry). This raises serious concerns, as no type of pollutant or contaminant should become a buzzword that overshadows other forms of deadly pollutants. Prioritization is one thing, but this can be deemed as ignorance that directly disrespects and further harms other victims. When Brock meets with the media, their first question is directed toward PFAS studies. Her stance is consistent, as PFAS is not her main concern, and with an exclusive focus on PFAS, policymakers are “missing the boat on the solvents and chemicals that military members work with” 7 . The current focus on PFAS, although important, obscures the broader chemical issue that continues to affect service members, their families, and surrounding communities. Brock’s observation that PFAS has been pushed as the only form of relevant pollution speaks directly to the need for a comprehensive approach that addresses a full range of hazardous substances released over the decades.

Additionally, the institutional response Brock has encountered shows the same patterns of delay, denial, and information control that are the main characteristics across military contamination cases. She described reading Veterans Administration disability claims that became public knowledge and being struck by “how many people are denied service connection for their illnesses, when there are studies out there that actually say those illnesses are linked to occupational chemical exposures.” Brock stated the Agency for Toxic Substances and Disease Registry provides fact sheets on each toxic chemical that individuals work with, describing the illnesses these men and women are suffering. Yet veterans continue to be denied disability benefits for conditions clearly due to their occupational exposure. From Brock's perspective, this pattern of denial reflects a cost issue. Acknowledging the link between service and these illnesses would require the government to provide disability compensation to tens of thousands of veterans and their survivors, which is a fiscal liability the DOD and Veterans Administration seem to be unwilling to accept.

Brock's response to claims that cancer among service members is attributable to genetics or familial habits is that her husband had no family history of cancer and never smoked. His best friend, who died from pancreatic cancer three months later, also had no family history and never smoked. The insistence on attributing service members' cancers to individual behaviors and genetic predispositions shifts the responsibility away from the institutions whose practices created the exposure. As Brock stated, genetics is a relatively new field, and the military has never conducted systematic studies about genetic histories or controlling for these variables in exposure assessments. The lack of such studies is, in itself, an institutional choice. It allows the government to continue claiming insufficient evidence, even as the human toll rises.

When asked whether she believes the government withheld or minimized information about contamination risks, Brock stated she believes "they do withhold a lot of information." She discussed a pattern across studies she has read: "They all have the same conclusion, not enough information to support, or, you know, inconclusive." This consistent finding despite mounting evidence, she characterized as institutional unwillingness rather than genuine scientific uncertainty, as is her distrust of government-conducted studies: "I don't trust studies, especially performed by the government." Overall, Camp Lejeune establishes a pattern for military contamination and for the decades-long struggle for acknowledgment, accountability, and justice.

Partners in Pollution: Corporate Profit through Contracts with the Military

The PFAS contamination crisis is about more than the Department of Defense's contaminants leeching into and staying in the environment, and the concept of forever chemicals is not foreign, nor was the use of them a mere accident. The overarching justification for many things related to PFAS is scientific uncertainty, but the reality is that there is only certainty in calculated corporate decision-making, deliberate concealment, and the prioritization of profit over human life and the

ecosystems upon which it relies. The corporate giants that manufactured and profited from these forever chemicals, mainly 3M and DuPont, held knowledge of PFAS toxicity decades before the public or regulators were informed (Richter et al. 2025). Their actions follow a pattern of corporate behavior that is built on the strategies of the tobacco and fossil fuel industries. This includes manufacturing doubt, suppressing evidence, and externalizing deadly costs onto society while enjoying the profits (Cordner et al. 2025).

Concealed Knowledge

Corporate profit is a large motivator in the mass production of AFFF. 3M and DuPont acted on that profit motive through the deliberate suppression of evidence and knowledge. Internal documents that were uncovered through litigation initiated by attorney Robert Bilott, whose story is dramatized in the film *Dark Waters*, showed that DuPont and 3M were aware of the dangers of PFAS as early as the 1950s (Bilott 2019). Investigative journalist Mariah Blake's book from 2025, *They Poisoned the World: Life and Death in the Age of Forever Chemicals*, utilizes internal industry documents and extensive interviews to understand the full picture of corporate greed. Blake wrote that the companies created, used, and marketed harmful substances while hiding implications (Blake 2025, p.4). A 2025 study analyzed industry documents that spanned from 1961 to 2006, and found "clear evidence that the chemical industry knew about the dangers of PFAS and failed to let the public, regulators, and even their own employees know the risks" (Richter et al. 2025, p.3). The documents detailed DuPont's pre-existing knowledge of enlarged livers in animals exposed to C-8 (PFOA), factory workers bearing children with birth defects, and animals dying within days of ingesting these chemicals. None of those findings were reported to the Environmental Protection Agency, which was required by law (Richter et al. 2025).

The timeline of knowledge is undeniable in proving corporate intent. Interestingly, the U.S. government approached DuPont to mass-produce

PFAS during World War II, but initially declined due to liability concerns. The government then agreed to indemnify DuPont, meaning the government would hold the responsibility for harm caused (Blake 2025, p. 78). By the 1960s, tests conducted by both DuPont and 3M revealed toxic effects of PFAS exposure on animal subjects, which included damage to the kidney, liver, and spleen (Bilott 2019). In 1978, 3M informed DuPont that it had confirmed the presence of PFAS in the blood of its own workers. A 1979 DuPont study found that PFAS bind to blood protein instead of breaking down and exiting the body, making the chemical biopersistent. In that same year, an internal survey of DuPont employees at a Teflon plant found increased cases of liver damage, but the exposed workers were not notified of those findings (Richter et al. 2025). By 1980, both companies reassigned female workers from PFAS-related jobs due to animal studies that indicate risks of fetal defects (Lerner 2022). In 1989, 3M's internal studies found increased levels of cancer in workers exposed to PFAS. And in 1992, DuPont confirmed those findings and documented increased cancer rates among its own employee population (Richter et al. 2025).

In Robert Bilott's memoir of his 20-year-long battle against DuPont, he wrote that when he first requested documents related to DuPont's PFOA emissions, the company provided a single box of papers. After months of legal pressure, the company produced "hundreds of thousands of pages of documents dating back decades", proving the company was aware of PFOA's potential danger since the early 1960s (Bilott 2019, p.87). The documents provided included internal studies showing that PFOA caused birth defects in rat offspring, accumulated in human blood, and remained in the body indefinitely. One internal DuPont memo from 1970 blatantly stated that PFOA "is toxic to humans and the environment and should be handled with extreme care" (Blake 2025, p. 156). However, these findings were never shared with nearby residents, workers, or regulators.

Disregarding the accumulating evidence, DuPont publicly insisted that its products were safe. In 1998 and 2002, DuPont emailed the EPA asking it “to quickly (like first thing tomorrow) say the following: That consumer products sold under the Teflon brand are safe and to date, there are no human health effects known to be caused by PFOA” (Lerner 2022, p. 145). This request was made at the same time its internal company documents demonstrated knowledge that PFOA had severe health risks, which is why Lerner terms “the playbook of deception,” in reference to the same tactics used by the tobacco industry to create doubt about the link between smoking and cancer (Lerner 2022, p. 89).

The most blatant example of corporate deception in the PFAS crisis is DuPont’s divestment of its environmental liabilities. In 2015, DuPont put its PFAS operations under a separate company, Chemours, transferring all PFAS-related environmental liabilities to the new entity (Stokes 2023). The separation agreement included provisions that explicitly moved to prevent Chemours from having any recourse against DuPont (New Jersey Attorney General 2022). DuPont’s strategic move protected its assets from the growing liabilities strongly linked to decades of PFAS contamination. However, DuPont’s estimations of liability were far too low, proven by its estimated liability for the Fayetteville Works facility in North Carolina at \$2 million, but in 2019, the settlement was over \$200 million for remediation of water contaminated by that facility. New Jersey, Ohio, New Hampshire, and Vermont joined in commencing legal actions, alleging that DuPont had committed fraud in divesting its PFAS liability by being intentionally designed for bankruptcy (Stokes 2023; New Jersey Attorney General 2022).

The actions of 3M, DuPont, and other PFAS manufacturers represent a systemic pattern where corporate actors take positions, ranging from resistance and stonewalling to strategic adaptation (Bartley & Fairbrother 2024, p. 12).

In the case of PFAS, the most popular corporate strategy is regrettable substitution, where “narrow bans and opaque/guarded safety information lead to the proliferation of alternatives that may not be safer” (Bartley & Fairbrother 2024, p. 15). The capitalist growth that drives this is not incidental; it is fundamental. Such an incentive structure operates through the pressure to maintain market share, the financial benefits of delaying regulation, the fiscal savings in avoiding toxicity testing, and the ability to place cleanup and health costs onto taxpayers (Bartley & Fairbrother 2024). So, even when companies adopt voluntary restrictions, they are avoiding a radical phase-out of PFAS (Karlsson et al. 2025, p. 8). Voluntary corporate accountability mechanisms appear progressive, but result in minimal compliance lacking substantive change (Karlsson et al. 2025, p. 12).

Although corporate strategies have historically alleviated polluters of accountability, these methods have begun to fail in the modern court of law. In 2023, 3M was pushed to an agreement to pay between \$10.5 billion and \$12.5 billion to public water systems across the U.S. to remediate PFAS contamination. Another agreement allocated up to \$4 billion in liability, with Chemours assuming 50 percent and DuPont and Corteva (another DuPont spin-off) dividing the remaining 50 percent (Stokes 2023). Most excitingly, in 2025, the Miteni case in Italy is the first time a national court has imposed criminal liability with substantial criminal sentences on corporate executives for PFAS-related environmental harm. The Court of Assizes of Vicenza convicted eleven former managers of the chemical company, imposing a total of 141 years in prison, while also awarding €56.8 million in damages to the Italian Ministry of the Environment (De Toni & Piva 2025).

What Does Corporate Behavior Reveal About Our System?

The documentation of concealed evidence, strategic liability shifting, and the substitution of one harmful chemical for another reveals foundational truths about the American regulatory system and the position of corporations within it.

Firstly, it speaks about the chemical regulatory framework. Specifically, the Toxic Substances Control Act (TSCA) has historically allowed chemicals to enter commerce without sufficient toxicity and exposure data, which creates what scholars term “undone science” or “structurally produced knowledge gaps” (Cordner et al. 2025; Richter et al. 2025). Secondly, for regulatory agencies, through statutory limitations or self-imposed reticence, they have failed to properly exercise their authority to mandate testing or restrict hazardous substances (Bartley & Fairbrother 2024).

Most fundamentally, the case of PFAS demonstrates that the current system does not just permit, but structurally incentivizes the kind of behavior DuPont and 3M exhibited. The profit motive to maximize shareholder returns creates a powerful pressure to minimize costs, including the costs of environmental protection, toxicity testing, and liability (Cordner et al. 2025). When companies can externalize such costs onto the families who lose loved ones to cancer, onto the communities of color and poverty whose water supplies are poisoned, and onto the taxpayers who ultimately fund the cleanup, they face no genuine penalty for doing so. Not only do they face no punishment, but the market incentive is so strong that corporations have purchased all of the seats at the table.

How Does Corporate Money Buy Political Power and Protection?

The primary manufacturers of PFAS, 3M and DuPont, have spent decades and hundreds of thousands to obtain political influence through campaign contributions, lobbying, and the placement of former industry executives in regulatory positions. The result of this is a system that rewards pollution. A system, as Pat Elder stated, that is “not an accident”⁸. The scale of corporate political spending is extravagant, as according to data compiled by the Environmental Working Group, the chemical industry spent more than \$110 million on federal lobbying in 2023. 3M and DuPont were top spenders, as spending translates into political access and legislative outcomes (EWG 2024).

When Congress considered the PFAS Action Act of 2021, the most comprehensive PFAS action bill to date, which would have designated PFOA and PFOs as hazardous substances under CERCLA, required the EPA to set drinking water standards, and allocated cleanup funding, the chemical industry ran an aggressive opposition campaign. The bill passed the House, but in the Senate, many politicians received substantial campaign contributions from PFAS manufacturers, thus preventing its advancement (Stokes 2023).

Moreover, former 3M and DuPont executives have held positions at the Environmental Protection Agency, the Department of Defense, and other agencies responsible for PFAS regulation and remediation. The current EPA administration has actively weakened PFAS protections while claiming progress, along with political appointees silencing career scientists and burying scientific findings (Southerland 2026). Regulatory capture, where the regulated entities control the regulators, explains why the EPA has delayed action for decades, suppressed toxicity assessments, and sought to rescind drinking water standards even after finalization. As Elder stated, “when you say EPA in the context of PFAS found at the Navy base, the EPA can hardly be considered as an independent agency” 9 .

The consequences of such political capture are human lives. When Congress passed the PACT Act in 2022, it was seen as a landmark expansion of benefits for veterans exposed to burn pits and other toxins. However, through Doris Brock’s observation, the final legislation fell short, as the government and VA “decided to only pick and choose which diseases or illnesses that they would cover,” which stripped out conditions that had been included, like male breast cancer 10 . The narrowing of a legislative scope after passing indicates the ongoing influence of lobbyists who work to limit the imposition of accountability. “Small pieces are going to cost the lives of tens of thousands of military members,” Brock noted, “I just think that that is a way the government can stall” 11 .

Delay saves money, and for as long as the system rewards delay through contributions and future industry employment, there is no fiscal reason to change. Despite the promise of representation, the seats where policy is made are occupied and purchased by the industry.

Preexisting Regulation and Precedent

Understanding the PFAS contamination crisis requires examining not only what the corporations did, but also how the system allowed them to do it. As environmental advocate and journalist Pat Elder put it, “The entire process is, I would go as far as to say criminal in nature,” referencing how the Navy serves “as the polluter but they’re also in charge of the cleanup and then they’re also in charge of deciding what to clean up and how to present that data to us” 12 . The arrangement of an entity being responsible for pollution simultaneously controlling the investigation, remediation, and communication of that pollution is a foundational governance failure.

Protector to Barrier

The Environmental Protection Agency (EPA) was established in 1970 during a time of bipartisan environmental consensus, created to serve as an independent entity capable of holding polluters accountable. As Elder reflected on his early career working with staffers connected to Senator Edmund Muskie of Maine (who was instrumental in the Clean Water Act and CERCLA), there was much excitement for the possibilities of progress available 13 . Although the formative years of the EPA were genuine, Elder also now states, “everything has kind of gone to hell” 14 . Dr. Betsy Southerland, who worked at the agency for over three decades, writes along similar lines as Elder. She discussed how the EPA under the Trump administration is gaslighting the public about PFAS contamination, claiming major progress while “actively weakening the very protections meant to reduce exposure to these toxic ‘forever chemicals’” (Southerland 2026).

The EPA’s most recent actions on PFAS show regulatory capture and intentional delay.

The Biden administration was giving water systems five years to comply with new PFAS drinking water standards for six chemicals, but the current EPA administrator, Lee Zeldin, is asking for four of those standards to be rescinded, along with plans to delay compliance for an additional two years. This request was rejected by federal courts, citing the lack of legal and scientific reasoning (Southerland 2026). Moreover, the EPA has delayed TSCA reporting requirements that required PFA manufacturers and importers to report uses, production volumes, exposures, hazards, and disposal from 2011 to 2022. Administrator Zeldin has proposed delaying that reporting by a year and exempting all importers of PFAS-containing articles (Southerland 2026). A main concern lies in the EPA suppressing scientific findings. The agency hid an updated PFAS toxicity assessment and eliminated the Office of Research and Development; “suppressing scientific findings does not make drinking water safer; it leaves people exposed” (Southerland 2026).

Perfect on Paper: CERCLA and Superfund Sites

CERCLA’s application to military installations has been largely compromised due to exceptions and exemptions. The Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA, also called Superfund), was enacted in 1980 to address the nation’s most hazardous waste sites. The statute establishes a framework for identifying contaminated sites, creating fiscal responsibility, and protecting communities from industrial contamination (Pollack et al. 2023). Substances designated as hazardous under CERCLA are subject to reporting requirements, specific spill rules, release tracking requirements, and additional reporting under different environmental statutes.

In theory, CERCLA represented a powerful tool for environmental justice and corporate accountability. As Pietkoski (2020) discusses, military facilities operate under exemptions, delayed compliance processes, or discretionary enforcement provisions, which allow contamination to continue without effective or appropriate remediation.

Instead, military sites operate through negotiated Federal Facility Agreements (FFAs) that prioritize smooth operations over quick remediation. These agreements are negotiated between the EPA, state environmental agencies, and the DOD, granting the military significant control over the pace and scope of investigation and/or remediation (Pietkoski 2020). As previously stated by Activist Pat Elder, a single entity having complete control over the consequences of its own actions is not effective.

State-Level Legislative Action

As federal action has stalled due to increasing political polarization, states have become a primary driver of PFAS regulation. As Elder noted, when evaluating PFAS health risks, the state of California created Proposition 65, where they designate PFOS and PFOA as human carcinogens ¹⁵. California's Proposition 65, enacted in 1986, mandates businesses to provide warnings about exposure to chemicals that cause cancer, birth defects, or any reproductive harm, with PFOA and PFOS on the long list of carcinogens.

Several other states have enacted comprehensive PFAS bans on consumer products. Colorado's SB 24-081 is the Perfluoroalkyl and Polyfluoroalkyl Chemicals Protection Act, banning the sale or distribution of cleaning products, cookware, dental floss, menstruation products, and ski wax containing intentionally added PFAS, as well as prohibiting the installation of artificial turf containing PFAS (Shook, Hardy & Bacon 2025). Connecticut's SB 292 prohibits the manufacture or sale of products with intentionally added PFAS unless manufacturers provide prior written notification to the state, and additional labeling requirements for outdoor apparel, turnout gear, and consumer products that have taken effect in 2026 (Shook, Hardy & Bacon 2025).

Maine's SP610 prohibits the sale or distribution of cleaning products, cookware, cosmetics, dental floss, juvenile products, menstruation products, textiles, ski wax, upholstered furniture, and HVAC equipment containing intentionally added PFAS, with one exception for what the

Department determines are “currently unavoidable.” This is a high bar that requires demonstrating the use is “essential for health, safety, or the functioning of society and for which alternatives are not reasonably available” (Shook, Hardy & Bacon 2025).

Minnesota’s Amara’s Law, named after a young girl affected by PFAS, requires manufacturers of all products containing intentionally added PFAS to submit detailed information to the state, with the inclusion of product descriptions, the purpose of PFAS use, and the amount of each PFAS present (Shook, Hardy & Bacon 2025). Those who fail to comply may be banned from selling the product in Minnesota entirely. The state of Maryland has also taken action with the Protecting State Waters from PFAS Pollution Act, which restricts PFAS discharges and increases reporting requirements. Federal facilities, including the NAS PAX base, remain subject to federal law rather than state law, creating a two-tiered regulatory system with less stringent standards for military installations (Handson 2017). As federal action has stalled, states across the country have enacted their own regulations. Although these various states are not geographically connected to PAX or Camp Lejeune, they represent a strong trend that advocates for the restriction and removal of PFAS. This signals that reform and change are possible, and that the two-tiered system offers hope.

SCOTUS Ruling: A Loss for State Jurisdiction

States nationwide in the U.S. have been seeking to remove PFAS from their markets. The two-tiered system of federalism grants states the power to do so, but federal preemption can overrule said power through Supreme Court rulings and federal decisions. On March 2, 2026, the U.S. Supreme Court denied certiorari to an appeal from Maryland and South Carolina, allowing the Fourth Circuit ruling to stand, which requires the states’ PFAS contamination cases against 3M to be heard in federal court (Taft PFAS Insights 2026; Maryland Daily Record 2026). There are four separate lawsuits filed by the states to recover damages for PFAS contamination.

The states strategically split their lawsuits by filing one suit for PFAS contamination generally, and a separate suit for AFFF contamination specifically. This was an effort to keep the general PFAS suits in state court (Maryland Daily Record 2026). The Fourth Circuit’s ruling held that 3M produced some PFAS materials at the direction of the federal government for military use, thus allowing the company to invoke the “federal officer removal statute” (28 U.S.C. § 1442(a)(1)), pushing the cases to be heard in federal court (Tafts PFAS Insights 2026). 3M argued that the military used MilSpec AFFF across the country, including both states in the suit. The company also argued that the plaintiff’s ability to identify whether the PFAS came from MilSpec AFFF or non-MilSpec AFFF is weak, drawing on the idea of scientific uncertainty. Their reasoning allows 3M to argue that the states’ general PFAS suits were related to its federal contractor work, ultimately triggering federal jurisdiction. This ruling is significant because federal courts are less favorable to plaintiffs in environmental cases due to stricter pleading standards, more limited discovery, and a history of dismissing claims against federal contractors under government contractor immunity doctrines. For states like Maryland and South Carolina, the U.S.S.C decision means their cases proceed in federal court, where corporate defendants have a better track record.

Sovereign Immunity and the Limits of Litigation

The most powerful legal barrier to accountability for military contamination is the doctrine of sovereign immunity, holding that the federal government cannot be sued without its consent. Although the Federal Tort Claims Act (FTCA) waives sovereign immunity for certain claims, it has specific exceptions that are interpreted to prohibit most military environmental contamination lawsuits (Stokes 2023). As Elder recounted from his advocacy work with survivors of contamination at George Air Force Base: “They sued, and the Air Force claimed sovereign immunity in federal court... and then Wurtsmith Air Force Base in Michigan sued, and again, sovereign immunity, which means we have the right, in the name of national security, to poison you” 16 .

In addition to the doctrine of sovereign immunity, the Feres doctrine, established through a Supreme Court ruling in 1950, restricts military members' ability to sue the government for injuries stemming from military service. Under this doctrine, members cannot bring FTCA claims against the government for injuries "incident to service", which is a category that courts interpreted broadly to include environmental exposure at military installations (Stokes 2023). The Camp Lejeune Justice Act of 2022 created an exception to sovereign immunity for individuals exposed to contaminated water at Camp Lejeune between 1953 and 1987, allowing individuals to file claims for injuries resulting from exposure. However, it has a requirement of exhausting administrative remedies before filing suit, imposed strict deadlines, and capped attorneys' fees, reducing the accessibility and effectiveness of its supposed goal (Stokes 2023).

Illusion of Participation: The Restoration Advisory Board System

The Restoration Advisory Board (RAB) was established in the 1990s as a response to community activism demanding greater transparency (Pietkoski 2020). While RABs represent a theoretical partnership between the military, regulators, and affected communities, RABs are advisory only. There is no decision-making authority, and the military retains final control over all cleanup decisions. The PAX individual interviewed honed in on the base's transparency, and when asked about independent testing, they responded that they did not see the value of third-party verification 17 . M. Dibbles stated that this response is "anti-scientific" and that "the foundation of scientific review itself is to peer review and check and verify outside of your internal constituencies" 18 . Elder's experience with the MDE study shows how independent testing can be of use at times. When the state tested for PFAS in the St. Mary's River, they set detection thresholds at parts per billion rather than parts per trillion, which is 1000 times higher than appropriate. A result of no concern for additional investigation is drastically different from independent testing resulting in PFAS at levels as low as 1.7 parts per trillion.

The RAB system, designed to provide community input on military environmental cleanup, functions as co-opting community concerns rather than incorporating them. M. Dibbles noted that true accountability would require “a proactive disclosure system... as soon as a contaminant is detected, and even before a contaminant is detected, working with the community directly and the surrounding communities with very transparent communication” 19 . This would build upon community-led monitoring with independent verification, which is vastly different from the current system, where the polluter controls both the investigation and communication of the results.

In addition to this, any citizen can request information from/about federal agencies through the Freedom of Information Act. Although designed for the sake of transparency, there are nine exceptions to a FOIA, including national security, personal privacy, and trade secrets. If denied, an individual has the right to file an administrative appeal. It is a helpful initiative and useful resource to those governed, but there are several flaws, as those nine exceptions are quite broad, and plenty of information excluded from a FOIA contains information that, morally speaking, should be available to the public. Only 12.1 percent of FOI requests were fully granted in 2024, and there are over 200,000 requests backlogged for the third consecutive year (Massoglia 2026). A lack of response is to be treated as an automatic constructive denial. The process of requesting information is obscure or unknown to the average citizen, and the specificity of the contents in each agency can be nonexistent, meaning that individuals cannot properly utilize this resource if there is not an adequate understanding of an agency and its particular content. Moreover, FOIAs legally require agencies to respond to a request within 20 business days, but there is no reinforcement of this, resulting in all agencies violating this rule. Request wait times vary by agency, but many take two or more months to respond, and some will never respond (constructive denial). There is also a potential fee attached to FOIAs, which makes the process inaccessible to those of certain economic backgrounds.

Through personal experience, a FOIA was filed with the Department of the Navy in mid-December of 2026 and was answered in late January of 2026. The request was for the contracts between 3M and the Navy, as 3M has been the Navy's main supplier of AFFF and was the first company to develop a patented AFFF formula. The request could not be fulfilled due to the documents possibly being stored at the National Archives and Records Administration (NARA). After receiving this response, another FOIA was immediately filed (in January of 2026) to the specific subsection of NARA, the archives of the Navy, for the same material. By late March, there had yet to be a response, and this is most likely due to a silent denial. So, a Freedom of Information Act request is a brilliant idea on paper, but in reality, it is an empty promise. An initiative for transparency is absolutely futile when most information is not available to the public, the process is difficult, and it is placed behind a potential paywall. This is not a genuine promise of transparency, aligning with the pattern of “good in theory” structures that are ineffective in reality.

Environmental Justice: What Does PFAS Mean in the Context of Justice?

Scientific literature and empirical data demonstrate that PFAS contamination, similar to all forms of industrial pollution, disproportionately affects low-income communities, communities of color, and Indigenous populations. These communities have come to bear the health burdens, the economic costs of contamination, and the emotional toll of fighting for accountability. As M. Dibbles observed, the fundamental problem is that "the entire model moves the accountability from the contaminating party... onto the victims and the communities. The burden of proof for accountability should not be on the community members." 20 .

The concept of environmental justice came to be through grassroots movements in the 1980s when the documentation of the disproportionate siting of hazardous facilities in communities of color gained traction.

The 1987 report, *Toxic Wastes and Race in the United States*, found that race was the most significant predictor of hazardous waste facility locations, more significant than income, homeownership, or property values (Bullard 2000). Robert Bullard documented how African American communities in the South have been historically targeted for landfills, incinerators, and chemical plants because they lacked the political power to resist (Bullard 2000). Checker (2005) continued this analysis to understand the operation of “polluted promises,” which is the gap between government assurances of safety and the lived reality of contaminated communities. Checker documented how residents of a predominantly African American community in Georgia dealt with decades of contamination from an industrial facility while government agencies dismissed them, downplayed the health risks, and delayed action. She identified a pattern of community members bearing the burden of documenting contamination, fighting for acknowledgement, and suffering health consequences while institutions profit from delay. This pattern is recurring in PFAS-affected communities nationwide.

Peer-reviewed research has increasingly documented the burden of PFAS contamination on marginalized communities. A 2024 study analyzed PFAS concentrations in public water systems across the U.S. and found that communities with higher proportions of Black and Hispanic residents had significantly higher PFAS levels in their drinking water compared to predominantly white communities, even after controlling for income and other socioeconomic factors (Andrews et al. 2024). It was concluded that racial and ethnic disparities in PFAS exposure require specific intervention and attention (Andrews et al. 2024, p. 8). Moreover, a 2023 analysis found that communities of color are more likely to live near military installations and industrial facilities. It was documented that “Black Americans are 1.4 times more likely than white Americans to live within one mile of a PFAS-contaminated military base” (Environmental Working Group 2023). Such a disparity is a direct result of redlining, exclusionary zoning, and the intentional siting of hazardous facilities in communities without political power.

The PAX individual interviewed discussed that minority communities live in hazardous areas because the hazard was placed there before, lowering property values, coinciding with how lower economic classes are typically of color. M. Dibbles addressed that framing of "minority communities move into environmentally hazardous areas rather than turning minority communities into hazardous waste sites" 21 . Dibbles explained that this framing presents "a half-truth, as if it's the whole story." Even if market dynamics play a role, "the market itself was based on racist housing policies, redlining, discriminatory lending, and exclusionary zoning. So no matter how you look at it, I think it's a self-reinforcing cycle," rather than a which came first question 22 .

The discovery of PFAS contamination also imposes economic burdens on impacted communities, and these burdens fall disproportionately on those with fewer resources to combat it. Huang and Kumar (2021) document that PFAS discovery destabilizes municipal finances because local governments must bear the costs of testing, treatment infrastructure, and legal action. Such costs strain local budgets and can discourage aggressive investigation or disclosure, creating underreporting and inadequate recognition (Huang & Kumar 2021, p. 14). The issue of economic impacts extends to property values. A 2024 study found that homes located within one mile of a PFAS contamination site experienced property value declines of 5 to 15 percent, with the largest impacts in lower-income neighborhoods that have difficulty absorbing losses (Muehlenbachs et al. 2024). For families whose homes are their primary source of wealth, these declines can be detrimental.

An incredibly visible and relevant example of environmental justice in PFAS disposal is the landfill in Emelle, Alabama. Emelle is located in Sumter County, Alabama, which has one of the highest poverty rates and largest Black populations in the state. The landfill is operated by Waste Management and is the largest hazardous waste landfill in the United States (Tns 2023). As the PAX individual stated, describing where the base sends its PFAS-contaminated waste, "They [Waste

Management] said, [we're] already taking a bunch of hazardous waste stuff. We'll take yours'" 23 .

When asked about the factor of contamination hurting minority communities in Emelle, as PAX sent their own waste there, "As far as the impact upon minority community or any community around the landfill, I would be concerned if it was just a subtitle D that did not have a liner and did not have leachate treatment, I would absolutely be concerned." 24 . However, there should still be concern because lining and leachate treatment systems, especially those at Emelle, repeatedly violate regulations. In 2023, it was reported that over two dozen containers of hazardous waste, including PCBs, were rusting or damaged, with 16 reported as leaking, as well as a two-year-old uncleaned spill from a leachate tank (Tns 2023). M. Dibbles addressed this inadequacy of relying on liners for protection, stating it is a mistake to place full confidence in lining infrastructure, and that even the EPA has expressed concern about liners preventing all liquids from entering the ground, as there are many well-documented cases of chemical corrosion, temperature stress, and pipe collapse 25.

The Burden of Fighting

A defining feature of environmental justice is that the burden of fighting for accountability falls almost entirely on affected communities. As Pat Elder documented through his decades of environmental advocacy, it is community members, not government agencies or responsible corporations, who have to bear the costs of testing, the emotional weight and grief, and years of advocacy to achieve even minimal acknowledgement. Elder's experience at PAX aligns with this pattern as he has spent years collecting samples.

Elder's own advocacy at Patuxent River exemplifies this pattern: he has spent years collecting samples, filing Freedom of Information requests, publishing reports, and pushing state and federal agencies to acknowledge contamination that official studies systematically understate. The emotional and financial costs are immense. Elder

disclosed the expense of independent PFAS testing: a single test kit for a rockfish costs \$350, and testing for oyster, crab, and water samples costs multiple times that cost. Yet without independent testing, communities would solely rely on government studies that, as Elder documented in MDE's work, set detection thresholds 1,000 times higher than appropriate. The burden of producing the evidence to receive appropriate remediation falls on community members who are already bearing both the health and economic costs of contamination.

The personal toll of bearing the burden is evident in the stories and lives of those who have lost loved ones to military contamination. Doris Brock's experience of her husband's death from bladder cancer after 35 years of service, and experiencing 34 of their 66 friends' deaths, is not something to be taken lightly. As she described, "The disappointment was not being able to spend the rest of my life with him... after we retired, which is... horrible, and there's so many people that are in the same shoes" 26 . The reality that Brock had to document these deaths herself, to track the outcomes of her husband's colleagues when no government agency would do so, reveals the foundational injustice of a system that puts the burden of proof on victims instead of the polluters.

Reform or Transformation?

The literature revolving around PFAS contamination has documented and described a system where corporate polluters thrive, military installations are shielded from accountability, and the harm and burden will always fall on the average citizen. The evidence presented discusses PFAS as more than just a scientific or technical problem. The science and data behind the PFAS crisis have evolved rapidly and will continue to do so to the benefit of society, but its implications in policy, regulation, and justice are complex. The concept of reform in this existing framework is questioned, as the patterns documented at Camp Lejeune, PAX, and countless other bases show a system so fundamentally broken that it requires an entire transformation rather than just incremental adjustments.

DISCUSSION

The findings of this research show a crisis that goes further than the chemical properties of PFAS. Although the carbon-fluorine bond explains why these compounds are so persistent in the environment, the persistence of regulatory system failures requires a separate explanation. The results of the four interviews, in addition to the peer-reviewed literature and legal analysis, show that the understanding of PFAS toxicity has dramatically advanced, but governance and remediation efforts remain unaltered. Such a paralysis is not just a failure of the system; it may be a feature of it.

The Weaponization of Uncertainty

The weaponization of uncertainty in the PFAS crisis is best understood through the concept of “power/ignorance”, the idea that powerful institutions actively produce ignorance as a strategy of governance. In the context of PFAS, there is no lack of knowledge or presence of uncertainty, at least not to the extent that inaction is justified. As established by Michel Foucault, power and knowledge are interconnected. Power creates knowledge, and knowledge enables power (Foucault 1980). Andrew Mathews (2005) refined this through his study of forest fires in Mexico, where the state used fires as justification for complete control of forests, coining the term “power/ignorance”. The chemical industry and the DOD have produced scientific knowledge on toxicity and persistence, but they have also strategically released this knowledge to delay regulation, undermine its reliability, and maintain the status quo.

The official state testing conducted by MDE yielded results that were contradicted by two independent data sets. MDE’s no detection rate of concern is due to setting detection thresholds too high to capture contamination. Although technological limits existed in 2020, when the study was conducted, the agency has continued to stand by its initial results, disregarding scientifically valid evidence that shows detection of concern. There has also been suppression of scientific findings, as the EPA hid an updated PFAS toxicity assessment done by its own Office of

Research and Development in April of 2025 (Southerland 2026). There is also consistent invocation of inconclusive science as a justification for delay or denial, which is a pattern that individuals like Doris Brock and Jerry Ensimer personally identified across years of government-conducted studies and interactions.

Moreover, information is withheld through FOIA exceptions, with only 12.1 percent of FOIA requests fully granted in 2024 (Massoglia 2026). So, as Mathews stated, this deployment of ignorance is not a failure of the state or system; it is a strategy of governance. Through the claims that the science is uncertain, that more research is needed, and that causation has not been definitively proven, the DOD and EPA have delayed regulation, disability claims, and avoided overall liability while the average citizen pays the price. This dynamic was also identified by M. Dibbles through their observation that the EPA's use of needing more time and additional studies has been heard since the 1970s and 1980s²⁷, leading to the question of whether these hesitations hold genuine weight or are they just another form of regulatory capture and delay?

Bad Law and Supreme Inequality

The concept of “power/ignorance” functions within a systematically biased legal system. In *Bad Law*, Elie Mystal, a former litigator and legal analyst, builds a crucial foundation for understanding how this bias works. Many of the harmful laws in the U.S. were not created out of good intentions; they are statutes that were specifically designed to cause harm: “these laws do not come to us from on high; we write them, and we can and should unwrite them” (Mystal 2025). This has an important place in the PFAS crisis, as the laws that enable contamination to continue, like the Toxic Substances Control Act, the Federal Tort Claims Act's exceptions that bar military environmental lawsuits, and the federal officer removal statute that allows 3M to move state cases to federal courts, should not be mistaken for accidents or oversights. They are policies and statutes that were written, lobbied for, and passed by people who understood the harmful implications to the fullest extent.

While incredibly harmful, it is simply the result of a designed and desired function. These are not neutral procedures and processes. They are bad laws that “continue to perpetuate hate, racial bias, injustice, and inequality today” (Mystal 2025).

Moreover, these bad laws typically have popular and bipartisan support. For example, the 2005 Protection of Lawful Commerce in Arms Act (PLCAA), granting gun manufacturers near complete immunity from product liability lawsuits, was passed with bipartisan support. This matters in the context of PFAS because bad environmental laws expand beyond one party or ideology. They are the desired outcome of a political system where corporate power is systematically privileged over public health.

Laws have measurable consequences. In the context of PFAS, the bad laws include CERCLA’s Federal Facility Agreements that grant military control over cleanup, and the TSCA provisions that allowed PFAS to enter commerce without proper toxicity testing. The measurable consequences of these laws take form in a human cost, such as the 34 friends and husband Doris Brock lost far too soon, the 9-year-old girl, Janey, who died of leukemia from base contamination, and the communities that surround Emelle, Alabama, who live alongside the nation’s largest hazardous waste landfill.

In *Supreme Inequality* (Cohen 2020), it documents how the Supreme Court has ruled in favor of wealthy individuals and corporations over the past 50 years, creating a system that solidifies inequality. The judicial branch was supposedly designed to address and prohibit wrongdoings, but it appears to function beyond the idea of checks and balances, and instead wields extreme power to permanently reinforce improper decisions. The Court’s ruling in *Citizens United v. FEC* (2010), allowing unlimited corporate political spending, and its generously expansive interpretation of the Feres doctrine are not anomalies. They fit into a pattern of jurisprudence that prioritizes corporate power over individual rights and overall public health.

The Supreme Court’s ruling on March 2, 2026, denying certiorari to Maryland and South Carolina, effectively allowing PFAS contamination cases against 3M to be heard in federal court over state court, is a solid example of this pattern. The Fourth Circuit’s ruling that the Supreme Court declined to review held that, due to 3M producing some PFAS-containing materials at the direction of the federal government for military use, the company can invoke the “federal officer removal statute” (28 U.S.C. § 1442(a)(1)). However, in a dissent by the Fourth Circuit Judge Henry Floyd, he warned that the majority’s ruling risked pushing “lawsuits properly before state courts” into the federal system, which moves “toward foreclosing state courtrooms to plaintiffs” (*State of Maryland V. 3M Company*, No. 24-1218 (4th Cir. 2025) 2025). By declining to review this ruling, the Supreme Court allowed a precedent that makes it substantially harder for states to hold PFAS manufacturers accountable, which is a decision that is consistent with the pattern described in *Supreme Inequality*.

The Accumulation of Disadvantage

Institutional and legal failures produce compounding harm for marginalized communities and can be characterized as cascading inequalities. A single environmental or societal disadvantage, like living near a military installation or hazardous waste site, being a person of color, or having a low income, does not exist in isolation. Rather, disadvantages accumulate and reinforce one another to produce outcomes that are worse than individual disadvantages. Through the lens of environmental racism, for a low-income family of color living near Naval Air Station Patuxent River, the cascading inequalities include proximity to contamination. Due to historical redlining and exclusionary zoning, a family of color lives in an area with higher PFAS levels in water, soil, and air (Andrews et al. 2024; Environmental Working Group 2023). In addition, the family would struggle to afford bottled water, home filtration systems, or the option of relocation. There would also be reduced property values, so their home, likely the family’s primary source of generational wealth, would lose 5 to 15 percent of its value due to PFAS contamination (Muehlenbachs et al. 2024). State and

federal testing protocols understate contamination, denying the family information necessary to make informed decisions. The family would also face increased risks for cancer, thyroid disease, immune suppression, and developmental effects (ITRC 2025). The family's ability to sue for damages would be reduced due to sovereign immunity, the Feres doctrine, and the federal officer removal statute (Stokes 2023). Lastly, due to political marginalization, the chemical industry's campaign contributions and lobbying ensure that legislation like the PFAS Action Act is stalled and narrowed.

Each of the disadvantages compounds upon another. The family cannot sue, cannot afford to relocate, cannot trust the government's testing results, and has no political power to change the system. The landfill in Emelle, Alabama, where PAX sent PFAS-contaminated soils, is the largest hazardous waste facility in the U.S., while also being located in a county with one of the highest poverty rates and the largest Black populations in the state. This is a network of cascading inequalities that goes as far as from Maryland to Alabama, to the contaminated homes and beings of marginalized communities across the nation. Lastly, in a scientific study mentioned previously, PFAS exposure does increase the chances of PCOS (polycystic ovary syndrome), a diagnosis that is frequently dismissed due to medical racism. People of color are 20 to 30 percent more likely to experience medical misdiagnoses compared to white patients (Team 2024). Women are 66 percent more likely to receive an incorrect initial diagnosis than men; about 1 in 5 women report having their symptoms ignored, and 70 percent of women have experienced medical gaslighting (Peters & Peters 2025). This creates the issue of statistical accuracy in surveying contaminated communities due to misdiagnoses or lack of diagnoses, providing further evidence that inequalities continue to compound due to systematic biases and structural inequalities.

An Intentionally Ignorant System Creates Predictable Outcomes

The PFAS crisis begs the question: Is the system broken, or was it built this way? The evidence provided suggests that the patterns documented at Camp Lejeune, Patuxent River, and various other military

installations are not accidents or singular failures. They are predictable outcomes of a system that has been designed and reinforced through Supreme Court rulings, statutory frameworks, regulatory capture, and the strategic production of ignorance, all to prioritize corporate profit and military impunity over environmental justice and basic public health. As Pat Elder stated, the system is “not an accident” 28 . As M. Dibbles observed, “you can’t really have a functioning environmental regulatory body under authoritarianism” 29 . The issue of PFAS is a part of deeper structural problems in American governance and government. These problems will require new laws in addition to institutional reform, as well as the consideration of who bears the costs of industrial production.

FUTURE RESEARCH

The evidence provided demonstrates that PFAS contamination is more than a scientific and technological problem, as it is a systemic governance failure rooted in the structural inequalities of American environmental law. Although the extent of the issue can be paralyzing, parts of these findings point toward paths for continued research and advocacy. One avenue stemming from this work is the opportunity for additional and deeper engagement with the Naval Air Station Patuxent River through the Restoration Advisory Board. As the individual from PAX noted, the RAB also includes civilian community members and extended an invitation to attend future RAB meetings.

Moreover, the installation restoration project manager at PAX directly reached out to gauge interest in filling a vacant role on the board. Although temporary due to the time constraints, such an invitation represents a solid opportunity for St. Mary’s College of Maryland students and faculty to participate in the process of remediation. Attendance at RAB meetings would provide students with exposure to the Federal Facility Agreement process, the dynamics and importance of local community engagement and action, and it would allow the college to serve as an independent observer of the base’s claims about contamination levels and remediation progress. It would also begin to address the PAX interviewee’s concern about low public participation.

Due to the college's physical proximity to the base, the college community has a part in the outcome and in the responsibility to engage in change.

Q Methodology Continuation

The interviews conducted showed significant diversity in how different stakeholders understand the PFAS crisis. The PAX individual honed in on transparency, proactive action, and the validity of scientific uncertainty. Pat Elder discussed systematic corruption, intentional deception, and even criminal negligence. Doris Brock spoke on institutional denial, personal losses and burdens, and the limits of legislative routes. M. Dibbles gave a more structured analysis of environmental justice in relation to regulatory capture and the limits of the current system. These differing perspectives are fundamentally different ways of understanding the problem, its causes, and what the proper solution is. A Q methodology study would be useful in continuation of this work, as it is a research approach designed to study subjectivity by identifying and characterizing the different perspectives that exist within a group of people on a specific topic. Differing from the typical question that prompts agreement or disagreement, Q methodology lets individuals sort statements following their personal subjective importance, which ultimately reveals “subject positions” that might not have been previously apparent.

A Q methodology study on PFAS contamination at military sites could assist with identifying the range of perspectives held by different stakeholders, like military individuals, activists, impacted residents, regulators/policymakers, scientists, and industry representatives. It would also quantify the degree of consensus or disagreement on issues like the reliability and acceptability of current testing protocols, the credibility of government versus independent studies, the effectiveness of the RAB process, and the perceived fairness of current legal pathways of remediation. It could also put different perspectives into demographic variables like proximity to contamination, duration of exposure, institutional role or affiliation, and political orientation. Lastly, it would build an empirical basis to identify areas of potential agreement, which may be utilized to inform policy reform and community/overall engagement strategies.

Additional Contacts

The interviewees for this research provided contacts for additional experts and advocates whose work could strengthen any variation or continuation of this project. Beginning with scientific and medical experts, Nate Donley, the Environmental Health Science Director at the Center for Biological Diversity, has extensive experience on the health impacts of PFAS and various other contaminants. Donley's organization has been at the forefront of litigation and advocacy to push for EPA action on PFAS regulation. There is also Linda Birnbaum, the former Director of the National Institute of Environmental Health Sciences (NIEHS), who has been described as a pioneer in PFAS research. Birnbaum dealt with political pressure and opposition for her willingness to speak publicly on the risks of PFAS. Such a perspective from a scientist who led the federal agency responsible for monitoring environmental health would be useful for understanding the science of PFAS and the politics within federal research. For biomonitoring and exposure assessments, Sean Lynch of the Maryland Pesticide Education Network has done work that provided direct evidence of PFAS causing harm to the human body. Lynch's experience and access to biomonitoring data would be crucial for understanding and establishing the link between environmental contamination and human exposure. There is also Jared Hayes of the Environmental Working Group who works on PFAS policy and can access EWG's mapping of PFAS contamination sites across the country. The EWG has played a major role in bringing scientific research into public advocacy.

On the topic of scientific integrity, Kyla Bennett is a former EPA scientist who became a whistleblower by documenting agency failures on PFAS and other chemicals. Her perspective shows the pressures scientists deal with when attempting to speak the truth. Bennett's knowledge of EPA operations and processes, as well as her willingness to criticize a government agency, brings an important narrative to additional research. There is also Laura Dumais, who authored a commentary for the Public Employees for Environmental Responsibility (PEER), an organization that has been vital in supporting whistleblowers and

pushing back against agency failures. Dumais's work focused on PFAS contamination in the food supply, which is a path of exposure that Elder warned was more significant than drinking water. On a similar note, journalistic experts like Sharon Lerner are important, as she has broken many of the most significant PFAS contamination stories, which include the concealment of evidence and the military's role in contamination. Lerner's reporting work was directly mentioned in this research as the interviewee, Brock, noted that Lerner "was way ahead of everyone and still is" 30 . Her expertise in topics such as FOIA requests, understanding and accessing industry documents, and connecting scientific findings to policy outcomes would be extraordinarily helpful. Lastly, the inclusion of organizations like Earthwatch and the Natural Resources Defense Council would build institutional support and legal expertise.

To create a more inclusive participant cohort, Francis Gray, the Tribal Chair of the Piscataway Conoy Tribe, would be a crucial perspective. Gray has mentioned in a guest lecture at St. Mary's College of Maryland that PFAS is a frustrating issue that has changed cultural norms and typical practices. Native American communities face significant challenges to preserving their land and culture. The Piscataway Conoy Tribe has dealt with severe violence and oppression to the extent of forced migration. The tribe secured its first piece of land in 400 years in 2023, showing the significantly delayed reparations. The inclusion of Gary and the Piscataway Conoy Tribe is important, as Southern Maryland has a history that must be acknowledged, especially as historical injustices persist in the modern day. Understanding the place of indigenous communities within the PFAS crisis would provide invaluable knowledge for remediation efforts.

Final Remarks

The plan moving forward includes immediate action like attending RAB meetings, as well as long-term structural change like the repealment of bad laws, limiting corporate political spending, ending sovereign immunity for military pollution, and working toward the burden of proof

falling on polluters. The stories and perspectives included in this research, such as Elder's elevated serum levels, Brock's deceased husband and friends, and the Camp Lejeune families who fought for 40 years, are more than just evidence or markers of harm. These perspectives must be treated as a call to action. PFAS harms various types of communities and people, changing lives in drastic and unimaginable ways. This contamination requires efficient, expansive, and meaningful changes to our legislative bodies and choices; change that more accurately reflects the idea of representing the people instead of corporate or agency interest. Without such change, the toll of PFAS will continue to rise, a toll measured in human lives.

Notes

- [1] PAX Individual, interview by author, St. Mary's City, MD, February 6, 2026.
- [2] Individual, interview.
- [3] Individual, interview.
- [4] Individual, interview.
- [5] Doris Brock, interview by author, St. Mary's City, MD, February 10, 2026
- [6] *Id.* at 42
- [7] *Id.* at 42
- [8] Pat Elder, interviewed by author, St. Mary's City, MD, February 9, 2026.
- [9] *Id.* at 52.
- [10] *Id.* at 42.
- [11] *Id.* at 42.
- [12] *Id.* at 52.
- [13] *Id.* at 52.
- [14] *Id.* at 52.
- [15] *Id.* at 52.
- [16] *Id.* at 52.
- [17] Individual, interview.
- [18] M. Dibbles, interview by author, St. Mary's City, MD, February 23, 2026.
- [19] *Id.*

[20] *Id.* at 61

[21] *Id.* at 61

[22] *Id.* at 61

[23] Individual, interview.

[24] Individual, interview.

[25] *Id.* at 61

[26] *Id.* at 42

[27] *Id.* at 61

[28] *Id.* at 52

[29] *Id.* at 61

[30] *Id.* at 42

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